

Mohamed Ahmed

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Education

University of Minnesota

Expected May 2027

Bachelor of Science in Electrical Engineering

Relevant Coursework: Embedded Systems, Analog Electronics, Wireless Hardware Design, Control Systems, Power Electronics, Antenna Design

Experience

Michels Power

May 2026 – Aug 2026

Project Management Engineering Intern

- Supported electric utility projects through field verification, GIS records, engineering drawings, and technical documentation.
- Reviewed engineering drawings and as-built records for transmission and distribution projects.
- Developed a technical database tracking **1,200+ utility as-built records** for project documentation.

Club Experience

STRIVE Medical

September 2025 – Present

Electrical / Hardware Team Lead

- Lead electrical development for a **medical robotics platform** integrating embedded control, sensing, actuation, and custom electronics.
- Integrated **Teensy 4.1, ESP32, IMUs, 14-bit encoders, and motor controllers** across robotic subsystems.
- Developed **sensor and motor-control firmware** and interfaced embedded hardware over I2C, SPI, UART, and BLE.
- Designed a **4-layer PCBA** supporting embedded controllers, sensor interfaces, power distribution, and motor-drive hardware.
- Brought up and debugged embedded hardware using oscilloscopes and logic analyzers; led electrical development to **unanimous 1st Place** at the 2026 UMN MVP Challenge.

Projects

Four-Port Gigabit Ethernet Switch | FPGA, Verilog, Embedded Hardware

- Contributed to development of a **four-port Gigabit Ethernet switch** built around a Lattice ECP5 FPGA and four Gigabit Ethernet PHYs.
- Developed and integrated **Verilog RTL modules** supporting FPGA-based Ethernet functionality and hardware interfacing.
- Worked with **RGMII and MDIO** interfaces for FPGA-to-PHY communication, PHY configuration, and link management.
- Supported RTL verification and debugging through simulation, synthesis, and hardware-oriented test workflows.
- Contributed to a custom multi-layer PCB integrating the **FPGA, Ethernet PHYs, power rails, clocking, and external interfaces**.

Wearable PPG Monitoring System | ESP32-C3, FreeRTOS, Embedded C, Real-Time DSP

- Developed **ESP-IDF/FreeRTOS firmware** for an ESP32-C3 wearable integrating optical PPG sensing, a 24-bit ADC, IMU, and BLE.
- Implemented **interrupt-driven SPI acquisition** using ADC data-ready events for deterministic sensor sampling.
- Synchronized PPG and IMU acquisition for real-time motion analysis and biosignal processing.
- Implemented a **101-tap FIR decimation pipeline** and IMU-referenced NLMS processing for motion-artifact reduction.
- Developed **C and Python test software** to verify DSP precision, filter response, data acquisition, and embedded processing.

Skills

Embedded: Embedded C/C++, FreeRTOS, ESP-IDF, ESP32-C3, STM32, Teensy 4.1, Interrupts, ADC, PWM

Interfaces: SPI, I2C, UART, BLE, RGMII, MDIO, GPIO

FPGA / Digital: Verilog, FPGA, RTL, Lattice ECP5, Digital Logic, Ethernet

Hardware: PCB Design, Embedded Hardware, Sensor Integration, PCB Bring-Up, Analog Interfaces

Debugging & Test: Oscilloscope, Logic Analyzer, Multimeter, Hardware/Software Debugging, Root Cause Analysis

Programming & Tools: C/C++, Python, MATLAB, Verilog, Altium Designer, KiCad, Git, LTspice